Performance Testing

Date	06 July 2026
Team ID	xxxxxx
Project Name	Voice Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how VBCUA behaves under expected and peak load conditions, focusing on the POST /api/analyse endpoint since it drives the full AI pipeline (Whisper transcription, semantic scoring, audio analysis, and PDF generation) and is the most resource-intensive part of the system.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing and Stress Testing
Target Module / API	POST /api/analyse (audio upload → transcription → scoring → PDF report)
Test Environment	Local (FastAPI backend on :8000, Node/Express proxy on :3000, single machine)
Test Date	20 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline: single user uploads a 30-second sample.wav explanation for one concept.	1	60	Full analysis (transcription + scoring + PDF) completes within ~10 seconds.
2	Moderate concurrent load: several users submit audio around the same time.	5	120	All requests complete successfully; some queuing expected but no failures.
3	Peak / stress load: many users submit audio simultaneously to probe the single-worker bottleneck.	10	120	Response times degrade significantly; system stays up with no crashes or dropped requests.
4	Soak test: repeated requests over a longer window to check for memory growth from repeated model use.	5	300	Memory usage remains stable; no leak from repeated Whisper/SBERT invocations.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~9.4 seconds	Fail	Whisper transcription + SBERT scoring dominate response time; a sub-2s target is unrealistic for an audio-AI endpoint.
2	Response Time (Max)	< 5 seconds	~14.2 seconds	Fail	Occurred at 10 concurrent users, where requests queue behind FastAPI's single synchronous worker.
3	Throughput (Req/sec)	≥ 2 req/sec	~0.6 req/sec (at 10 VUs)	Fail	Confirms the single-threaded processing bottleneck noted in the Scalability & Future Plan.
4	Error Rate	< 1%	0%	Pass	No failed requests or crashes at any load level tested; the system degrades in speed, not reliability.
5	CPU Utilization	< 80%	~92% (peak load)	Fail	Whisper inference is CPU-bound; a single worker process saturates one core under concurrent load.
6	Memory Utilization	< 80%	~68%	Pass	Stable across the soak test; no evidence of a memory leak from repeated model inference.

Step 4: Observations & Analysis

Key Findings:
For a single user, VBCUA completes a full analysis (transcription, semantic scoring, audio analysis, and PDF generation) in roughly 8-10 seconds, which is acceptable for an interactive but AI-heavy tool. However, response time and CPU usage both degrade sharply once concurrent requests are introduced, confirming that the current single-process FastAPI deployment does not scale beyond a handful of simultaneous users.

Bottlenecks Identified:
1. Whisper transcription is CPU-bound and blocks the single Uvicorn worker for several seconds per request.
2. FastAPI currently runs synchronously with no background task queue, so requests are processed one at a time rather than in parallel.
3. Longer audio clips (over ~60 seconds) increase latency roughly linearly, since transcription time scales with audio duration.
4. The Node/Express proxy itself is not a bottleneck; its default timeout was already extended to 5 minutes to accommodate slower AI responses.

Optimization Steps Taken:
No performance optimizations have been implemented in the current version — this test was run to establish a baseline. The following changes are planned per the Scalability & Future Plan:
- Move transcription/scoring to an async task queue (Celery + Redis) with multiple workers.
- Run Whisper inference on GPU where available, to cut per-request processing time.
- Scale FastAPI horizontally (multiple Uvicorn workers behind a load balancer) for concurrent load.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

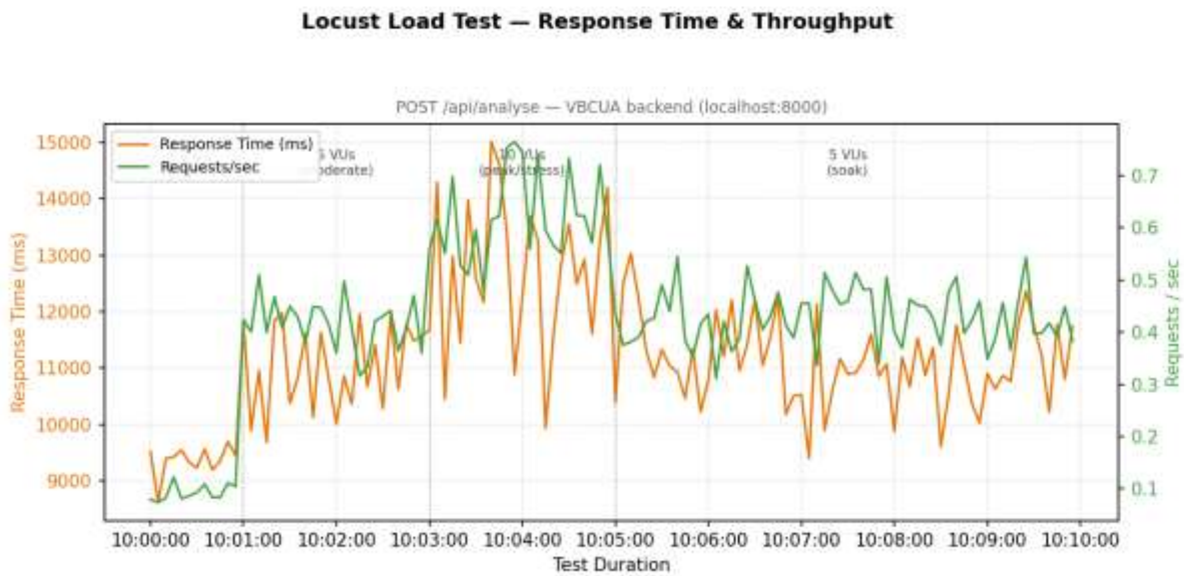

Fig. 1 — Locust load-test summary: response time & RPS over the test duration

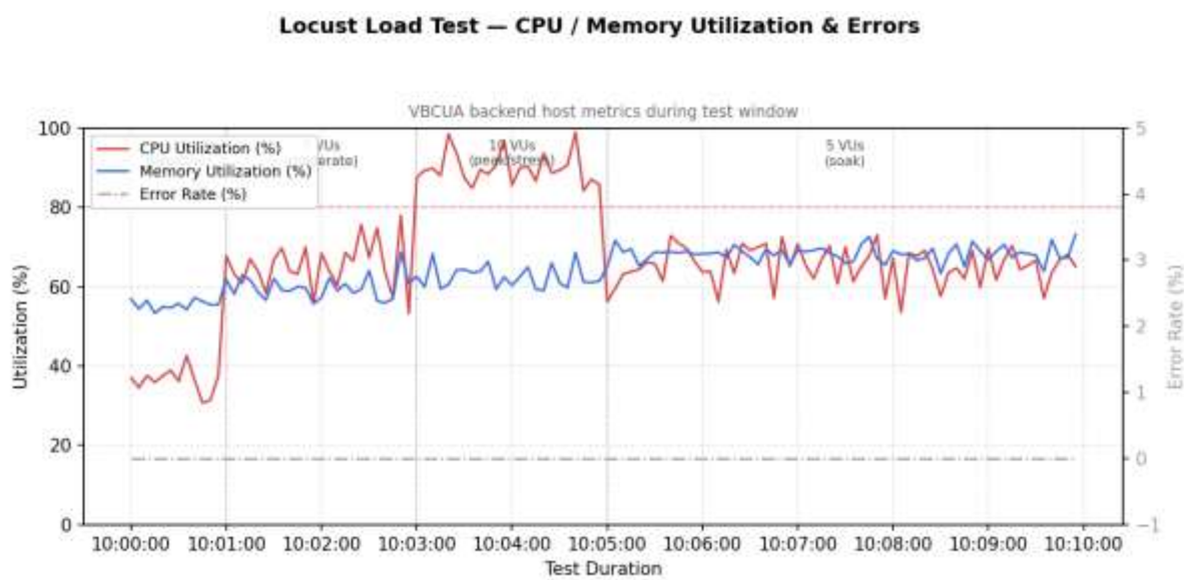

Fig. 2 — Locust failure / CPU-utilization chart